

(*) On every Riemannian manifold, we will always use the Levi-Civita connection if there is no further comment.

1 Some Constructions on Riemannian Manifolds

1.1 Flat Operators and Sharp Operators

Given a Riemannian metric g on the smooth manifold M , we define a bundle homomorphism

$$\hat{g} : TM \rightarrow T^*M$$

$$v \mapsto \hat{g}(v)$$

where $(\hat{g}(v))(w) := g_p(v, w)$ for any $v, w \in T_pM$. More over, if $X, Y \in \mathfrak{X}(M)$, this means $(\hat{g}(X))(Y) := g(X, Y)$.

Now, given a smooth local frame (E_i) for the tangent bundle TM and its dual coframe (ε^i) for T^*M , let $g = g_{ij}\varepsilon^i \otimes \varepsilon^j$ be the local expression for g . Then for $X = X^i E_i \in \mathfrak{X}(M)$, we have

$$\hat{g}(X) = (g_{ij}X^i)\varepsilon^j,$$

which means that the matrix of $\hat{g}$ in any local frame is the same as the matrix of g itself, i.e.,

$$\hat{g} \begin{pmatrix} E_1 \\ \vdots \\ E_n \end{pmatrix} = \begin{pmatrix} g_{11} & \dots & g_{1n} \\ \vdots & & \vdots \\ g_{n1} & \dots & g_{nn} \end{pmatrix} \begin{pmatrix} \varepsilon^1 \\ \vdots \\ \varepsilon^n \end{pmatrix}$$

now we denote

$$X_j := g_{ij}X^i$$

then we get

$$\hat{g}(X) = X_j \varepsilon^j$$

which is a standard practice to denote the components of $\hat{g}(X)$. We say that $\hat{g}(X)$ is obtained from X by **lowering an index**. We denote

$$X^\flat := \hat{g}(X).$$

Now, since the matrix (g_{ij}) is nonsingular at each point, the map $\hat{g}$ is in fact an isomorphism and thus invertible, by linear algebra we know that the matrix of $\hat{g}^{-1}$ is the inverse matrix of (g_{ij}) , which we will denote by (g^{ij}) , i.e., $g^{ij}g_{jk} = g_{kj}g^{ji} = \delta_k^i$. More over, the symmetry of (g_{ij}) implies the symmetry of (g^{ij}) .

Let the dual coframe (ε^i) be as above, for $\omega = \omega_j \varepsilon^j \in T^*M$, we have

$$\hat{g}^{-1}(\omega) = (g^{ij}\omega_j)E_i$$

which means that the matrix of $\hat{g}^{-1}$ is precisely (g^{ij}) (this can also be obtained from the knowledge of linear algebra), i.e.,

$$\hat{g}^{-1} \begin{pmatrix} \varepsilon^1 \\ \vdots \\ \varepsilon^n \end{pmatrix} = \begin{pmatrix} g^{11} & \dots & g^{1n} \\ \vdots & & \vdots \\ g^{n1} & \dots & g^{nn} \end{pmatrix} \begin{pmatrix} E_1 \\ \vdots \\ E_n \end{pmatrix}$$

now we denote

$$\omega^i := g^{ij}\omega_j$$

then we get

$$\hat{g}^{-1}(\omega) = \omega^i E_i$$

which is a standard practice to denote the components of $\hat{g}^{-1}(\omega)$. We say that $\hat{g}^{-1}(\omega)$ is obtained from ω by **raising an index**. We denote

$$\omega^\sharp := \hat{g}^{-1}(\omega).$$

These two isomorphisms are called **musical isomorphism**. The operators $\flat$ and $\sharp$ are called **flat operator** and **sharp operator** respectively.

1.2 Flat Operators and Sharp Operators on Tensors Fields

The flat operator $\flat$ and the sharp operator $\sharp$ discussed above can be applied to tensor fields of any rank on the Riemannian manifold (M, g) , in any index position, to convert tensors from covariant to contravariant or vice versa.

For $F \in \Gamma(T^{(k,l)}(TM))$, define

$$\sharp : \Gamma(T^{(k,l)}(TM)) \rightarrow \Gamma(T^{(k+1,l-1)}(TM))$$

$$F \rightarrow F^\sharp,$$

where $F^\sharp(\alpha_1, \dots, \alpha_{k+l}) := F(\alpha_1, \dots, \alpha_{i-1}, \alpha_i^\sharp, \alpha_{i+1}, \dots, \alpha_{k+l})$ whenever α_i are vectors or covectors as appropriate, here $i \in \{1, \dots, k+l\}$ is any covariant index position for F . In any local frame, the components of $F^\sharp$ are obtained by multiplying the components of F by g^{kl} and contracting one of the indices of g^{kl} with the i th index of F .

Similarly, we can define

$$\flat : \Gamma(T^{(k,l)}(TM)) \rightarrow \Gamma(T^{(k-1,l+1)}(TM))$$

by $F^\flat(\alpha_1, \dots, \alpha_{k+l}) := F(\alpha_1, \dots, \alpha_{i-1}, \alpha_i^\flat, \alpha_{i+1}, \dots, \alpha_{k+l})$, here i is a contravariant index position. In components, it is computed by multiplying by g_{kl} and contracting.

For example, let $A = A_{ij}^k \varepsilon^i \otimes \varepsilon^j \otimes E_k$ be a $(1, 2)$ -tensor field, then $A^\flat$ is a $(0, 3)$ -tensor field which is denoted by

$$A^\flat = A_{ijk} \varepsilon^i \otimes \varepsilon^j \otimes \varepsilon^k$$

where $A_{ijk} = g_{kl} A_{ij}^l$.

Another important application of the flat and sharp operators is to extend the trace operator on the tensor fields.

Recall: There is an operation called **trace** or **contraction** on the tensor fields on (M, g) , which is given by

$$\text{tr} : \Gamma(T^{(k,l)}(TM)) \rightarrow \Gamma(T^{(k-1,l-1)}(TM))$$

$$F \mapsto \text{tr} F,$$

here if F is expressed by

$$F \triangleq F_{j_1 j_2 \dots j_l}^{i_1 i_2 \dots i_k} E_{i_1} \otimes E_{i_2} \otimes \dots \otimes E_{i_k} \otimes \varepsilon^{j_1} \otimes \varepsilon^{j_2} \otimes \dots \otimes \varepsilon^{j_l}$$

then the tensor field $\text{tr} F$ is given in components by

$$\begin{aligned} \text{tr} F &\triangleq (\text{tr} F)_{j_1 j_2 \dots j_{l-1}}^{i_1 i_2 \dots i_{k-1}} E_{i_1} \otimes E_{i_2} \otimes \dots \otimes E_{i_{k-1}} \otimes \varepsilon^{j_1} \otimes \varepsilon^{j_2} \otimes \dots \otimes \varepsilon^{j_{l-1}} \\ &= F_{j_1 j_2 \dots j_{l-1} m}^{i_1 i_2 \dots i_{k-1} m} E_{i_1} \otimes E_{i_2} \otimes \dots \otimes E_{i_{k-1}} \otimes \varepsilon^{j_1} \otimes \varepsilon^{j_2} \otimes \dots \otimes \varepsilon^{j_{l-1}}. \end{aligned}$$

particularly, if $F \in \Gamma(T^{(1,1)}(TM))$, then $\text{tr} F = F_i^i$.

Now, if $h \in \Gamma(T^{(0,k)}(TM))$ with $k \geq 2$, i.e., h is a covariant k -tensor on the Riemannian manifold with $k \geq 2$, then we define the **trace of h with respect to g** as

$$\text{tr}_g(h) := \text{tr}(h^\sharp).$$

in particular, if $h \in \Gamma(T^{(0,2)}(TM))$, in this case $h^\sharp$ is a $(1,1)$ -tensor field and thus can be regarded as an endomorphism field. (recall that $L(V, V, \dots, V; V) \cong L(V^*, V, V, \dots, V; \mathbb{R})$) Hence $\text{tr}_g(h)$ is just the ordinary trace of this endomorphism field, i.e., $\text{tr}_g(h) = h_i^i = g^{ij} h_{ij}$ in terms of a basis, thus in particular, in an orthonormal frame $\text{tr}_g(h) = h_{ii}$ since $g^{ij} = \delta^{ij}$.

2 Ricci Curvature and Scalar Curvature

Ricci Curvature

Definition 2.1. : For a Riemannian manifold (M, g) , define a covariant 2-tensor field Ric as

$$\text{Ric}(X, Y) := \text{tr}(Z \mapsto R(Z, X)Y) = \text{tr}_g(Rm)$$

for any $X, Y \in \mathfrak{X}(M)$. It is called **Ricci curvature** or **Ricci tensor**.

Remark: In local frame, we write

$$\text{Ric} \triangleq R_{ij} \varepsilon^i \otimes \varepsilon^j$$

hence we have the equations

$$R_{ij} = R_{kij}^k = g^{km} R_{kijm}$$

where R_{ijk}^l and R_{ijkl} are the components of R and Rm respectively.

Scalar Curvature

Definition 2.2. : Define the **scalar curvature** on (M, g) to be the function S , which is given by

$$S := \text{tr}_g(\text{Ric}) = R_i^i = g^{ij} R_{ij}.$$

Remark: Through the Ricci curvature and scalar curvature on the n -dimensional Riemannian manifold (M, g) , we can define the **traceless Ricci tensor of g** as the following symmetric 2-tensor:

$$\mathring{\text{Ric}} := \text{Ric} - \frac{1}{n}Sg.$$

Proposition 2.1. : *Let (M, g) be a Riemannian manifold, then $\text{tr}_g(\mathring{\text{Ric}}) \equiv 0$.*

Proof: Note that

$$\text{tr}_g(g) = g^{ij}g_{ij} = \delta_i^i = n$$

hence

$$\text{tr}_g(\mathring{\text{Ric}}) = \text{tr}_g(\text{Ric}) - \text{tr}_g\left(\frac{1}{n}Sg\right) = 0.$$

which is the conclusion. $\square$

Exterior Covariant Derivative

Definition 2.3. : If $T \in \Gamma(T^{(0,2)}(TM))$, then define the **exterior covariant derivative** of T as

$$D : \Gamma(T^{(0,2)}(TM)) \rightarrow \Gamma(T^{(0,3)}(TM))$$

$$T \mapsto DT,$$

where $(DT)(X, Y, Z) := -(\nabla T)(X, Y, Z) + (\nabla T)(X, Z, Y)$.

Remark: In components, this means

$$(DT)_{ijk} = -T_{ij;k} + T_{ik;j}$$

here, we denote the components of ∇T as $\nabla T \triangleq T_{ij;k}\varepsilon^i \otimes \varepsilon^j \otimes \varepsilon^k$.

Recall: Recall that we have defined the **total covariant derivative** as

$$\nabla : \Gamma(T^{(k,l)}(TM)) \rightarrow \Gamma(T^{(k,l+1)}(TM))$$

$$F \mapsto \nabla F,$$

where

$$\nabla F(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l, X) := (\nabla_X F)(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l)$$

$$:= X(F(\omega^1, \dots, \omega^k, Y_1, \dots, Y_l))$$

$$- \sum_{i=1}^k F(\omega^1, \dots, \nabla_X \omega^i, \dots, \omega^k, Y_1, \dots, Y_l) - \sum_{j=1}^l F(\omega^1, \dots, \omega^k, Y_i, \dots, \nabla_X Y_j, \dots, Y_l)$$

for any $\omega^i \in \Omega^1(M)$ and any $Y_j \in \mathfrak{X}(M)$. Particularly, for $f \in C^\infty(M)$, we get $\nabla f = df$.

Proposition 2.2. : *The musical isomorphism commutes with the total covariant derivative operator, i.e., we have*

$$\nabla(F^\flat) = (\nabla F)^\flat$$

and

$$\nabla(G^\sharp) = (\nabla G)^\sharp$$

for any smooth tensor fields F and G on the Riemannian manifold (M, g) .

Proof: By the discussion in the previous section we can know that $F^\flat = \text{tr}(F \otimes g)$, note that g is parallel, i.e., $\nabla g \equiv 0$ since the Levi-Civita connection is compatible with g . So for $\forall X \in \mathfrak{X}(M)$, we get

$$\nabla_X(F \otimes g) = (\nabla_X F \otimes g) + (F \otimes \nabla_X g) = \nabla_X F \otimes g$$

since ∇_X commutes with the trace operation, so we get

$$\nabla_X(F^\flat) = \nabla_X(\text{tr}(F \otimes g)) = \text{tr}(\nabla_X F \otimes g) = (\nabla_X F)^\flat.$$

Now taking $F = G^\sharp$, since $\flat$ and $\sharp$ are inverse of each other, we get

$$\nabla_X(G^\sharp) = (\nabla_X G)^\sharp.$$

Since X is arbitrary, the conclusion holds. $\square$

Contracted Bianchi Identity

Proposition 2.3. : *Let (M, g) be a Riemannian manifold. The covariant derivatives of the Riemannian, Ricci and scalar curvature satisfy the following equalities:*

$$\text{tr}_g(\nabla Rm) = -D(\text{Ric})$$

$$\text{tr}_g(\nabla \text{Ric}) = \frac{1}{2}dS$$

in components, this is

$$R_{ijkl; \cdot}^{\cdot} = R_{jk; \cdot l} - R_{jl; \cdot k}$$

$$R_{il; \cdot}^{\cdot} = \frac{1}{2}S_{; \cdot l}$$

Proof: Note that by the differential Bianchi identity, we have

$$R_{ijkl; m} + R_{ijlm; k} + R_{ijmk; l} = 0$$

raising the index m and contracting the index i , we get the first equality in components. (use the conclusion in Proposition 2.2) Then do the same thing with the index j, k and simplify to obtain the second equality in components. (The details are direct computation) $\square$

3 Einstein Metric

Einstein Metric

Definition 3.1. : A Riemannian metric g is said to be an **Einstein metric** if its Ricci tensor is a constant multiple of the metric, i.e.,

$$\text{Ric} = \lambda g$$

for some constant λ .

The next proposition shows that it is not necessary to assume that λ is constant, just assuming that the Ricci tensor is a function times metric is sufficient.

Schur's Lemma

Proposition 3.1. : Suppose (M, g) is a connected Riemannian manifold of dimension $n \geq 3$ whose Ricci tensor satisfies $\text{Ric} = fg$ for some $f \in C^\infty(M)$. Then f is a constant and thus g is an Einstein metric.

Proof: Since $\text{Ric} = fg$ and $\text{tr}_g(g) = n$, so we have

$$\text{tr}_g(\text{Ric}) = S = \text{tr}_g(fg) = nf$$

which means that $f = \frac{1}{n}S$ and hence $\mathring{\text{Ric}} = \text{Ric} - \frac{1}{n}Sg = 0$. So $\nabla \mathring{\text{Ric}} = 0$, which implies that

$$R_{ij;k} - \frac{1}{n}S_{;k}g_{ij} = 0$$

in components, here note that $\nabla g = 0$. Now, tracing this equation on i and k , we get $g^{ik}R_{ij;k} - \frac{1}{n}S_{;k}g_{ij}g^{ik} = R_{ij;~}^i - \frac{1}{n}S_{;j} = 0$ (note that $g_{ij}g^{ik} = \delta_j^k$), by Proposition 2.3, we get

$$\frac{1}{2}S_{;j} - \frac{1}{n}S_{;j} = 0$$

because $n \geq 3$, we get $S_{;j} = 0$. Since $\nabla S = dS = S_{;j}\varepsilon^j = 0$, so by the connectedness of M , we know that S is a constant and thus so is $f = \frac{1}{n}S$. $\square$

Corollary 3.1. : If (M, g) is a connected Riemannian manifold with dimension $n \geq 3$, then g is an Einstein metric $\iff \mathring{\text{Ric}} = 0$.

Proof: ($\Rightarrow$): Let $\text{Ric} = \lambda g$, then we get $\lambda = \frac{1}{n}S$ by taking the trace w.r.t. g , hence $\mathring{\text{Ric}} = \text{Ric} - \frac{1}{n}Sg = \text{Ric} - \lambda g = 0$;

($\Leftarrow$): Since $\mathring{\text{Ric}} = \text{Ric} - \frac{1}{n}Sg = 0$, we have $\text{Ric} = \frac{1}{n}Sg$ and $\frac{1}{n}S \in C^\infty(M)$. So by Schur's Lemma, g is an Einstein metric. $\square$

参考文献

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